Human Computer Interaction

Designing the Mobile User Experience & Apple HIG

Comprehensive Class Notes

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1 Introduction to Mobile User Experience

Designing for mobile is fundamentally different from designing for desktop. The core distinction lies in the definition of "mobile" itself.

Core Concept: What is "Mobile"?

Fundamentally, **"mobile" refers to the user, not the device or the application.** The device is merely a tool that moves with the user through different contexts, environments, and states of attention.

1.1 The Carry Principle

The "Carry Principle" defines the nature of the **Personal Communication Device (PCD)**. A mobile phone is not just a small computer; it is an extension of the user.

- **Personal:** The device belongs to one person. It is personally identifiable (via accounts, biometric data, phone numbers). It creates an emotional bond with the user.
- **Communicative:** Its primary function is connection. It sends/receives messages and connects to networks constantly.
- **Handheld:** It is portable and operable with a single hand (thumb interaction). This imposes physical constraints (touch targets, screen reachability).
- **Wakable:** The device is "always on" or can be awakened instantly by the user or the network (notifications). This implies a state of constant readiness.

1.2 Implications of the Carry Principle

Because the device is always carried, designers must consider:

1. **Form Factor:** Size and weight matter.
2. **Features:** Cameras, sensors, and GPS are always available.
3. **Single Window Operation:** Unlike desktops with overlapping windows, mobile users typically focus on one full-screen task at a time (though split-screen exists, single-focus is dominant).
4. **Proliferation:** Apps must compete for attention on a crowded screen.

2 Characteristics of the Mobile User

Understanding the psychology and environment of the mobile user is crucial for UX design.

2.1 1. Interruptible and Easily Distracted

Mobile usage often occurs in bursts (micro-interactions) rather than long sessions.

- *Design Implication:* Interfaces must save state automatically. If a user looks up to catch a bus, they should be able to resume exactly where they left off without losing data.

2.2 2. Available

The user is reachable anywhere. This blurs the line between work and leisure.

2.3 3. Sociable

Mobiles facilitate "Groups and Tribes."

- **Voice Texting:** Immediate connection.
- **Extended Online Communities:** Social media on the go.
- **Physical/Mobile Hybrids:** Using a phone to interact with the physical world (e.g., scanning a QR code at a restaurant).
- **Status Symbol:** The device itself can be a fashion statement or status indicator.

2.4 4. Contextual

This is the most powerful differentiator. The device knows *where* it is and *what* is happening around it.

- **Sensors:** Accelerometers, temperature sensors, fingerprint readers, GPS.
- **Intelligent Behavior:**
  - A calendar app switching the phone to vibrate during a meeting.
  - A travel app alerting the user to leave early because of traffic conditions (combining GPS + Traffic Data + Flight Schedule).
  - A coupon app triggering a notification when the user walks past a specific restaurant.

Real-World Example: Contextual Awareness in Uber

Uber uses context extensively. It knows your location (GPS), the time of day (suggesting "Home" or "Work"), and real-time traffic data to calculate pricing and ETA. It doesn't ask the user "Where are you?"—it already knows.

2.5 5. Identifiable

The device acts as a digital ID card (SIM, Apple ID, Google Account), allowing for personalized experiences and secure transactions without constant login prompts.

3 Components of a Mobile Application

A mobile application is an ecosystem, not just code. It consists of:

1. **The PCD:** The hardware with its specific capabilities (battery, screen size).
2. **The User:** The human element, potentially distracted (riding a train, walking, eating).
3. **The Platform:** The OS (iOS/Android) and environment (browser vs. native).
4. **Output Interfaces:** Screen, speakers, haptics (vibration), Bluetooth audio.
5. **Input Interfaces:** Touchscreen, microphone (Siri/Voice), camera, GPS, Biometrics.
6. **Network/Carrier:** The infrastructure connecting the user to the internet.
7. **Server Infrastructure (Optional):** The backend cloud services syncing data.

4 Apple Human Interface Guidelines (HIG)

Apple's HIG provides the foundational principles for designing fluid, intuitive iOS apps. These principles are divided into three core themes and specific operational guidelines.

4.1 The Three Core Themes of iOS Design

4.1.1 1. Clarity

The interface must be intelligible and precise.

- **Legibility:** Text must be readable at every size. Use system fonts (San Francisco) to ensure readability.
- **Icons:** Icons should be precise and lucid.
- **Negative Space:** Use white space to let content breathe.
- **Color:** Let color simplify the UI (e.g., using blue only for interactive elements like links/buttons).

4.1.2 2. Deference

The UI should never compete with the content. The interface is the frame; the content is the artwork.

- **Fluidity:** The UI helps people interact with content but recedes into the background.
- **Borderless Buttons:** Use context and color (e.g., "Edit" in blue text) rather than heavy button borders.
- **Translucency:** Use translucent backgrounds (blur effects) to hint at content behind the current view, maintaining context.

4.1.3 3. Depth

iOS uses the Z-axis (layers) to communicate hierarchy.

- **Visual Layers:** Folders appearing to float above the home screen.
- **Motion/Transitions:** Realistic motion imparts vitality.
- **Zooming:** Example: The Calendar app. Tapping a year zooms into the month, tapping a month zooms into the day. This transition explains the user's navigational path intuitively.

Real-World Example: Depth in Control Center

When you swipe down the Control Center on an iPhone, the background blurs. This tells the user: "You haven't left your app; you are just looking at a temporary layer above it."

4.2 Detailed HIG Principles

In addition to the three themes, Apple outlines 8 specific guidelines:

1. **Clarity:** Clear language, icons, and layouts (as discussed above).
2. **Deference:** Content takes center stage.
3. **Depth:** Using shadows, blur, and motion to show spatial relationships.

4. **Feedback & Responsiveness:**
    - Every action needs a reaction.
    - **Haptics:** Small vibrations when a task completes.
    - **Visuals:** Buttons highlighting when pressed.
5. **Direct Manipulation:**
    - Users should feel like they are touching the content, not a control.
    - *Example:* Pinching a photo to zoom is direct manipulation. Clicking a "+" button is indirect manipulation.
    - Use standard gestures: Swipe, Drag, Pinch.
6. **Accessibility/Adaptability:**
    - Support **Dynamic Type** (users changing font size in settings).
    - Ensure distinct color contrast for visually impaired users.
    - Proper touch target sizes (minimum 44x44 points).
7. **Aesthetic Integrity:** The look must match the function. A banking app should look serious and secure; a game should look fun.
8. **Use of Standard Components:** Use native iOS pickers, alerts, and navigation bars. This ensures consistency and lowers the learning curve for users.

5 Applied HCI: NayaPay Case Study

The following class activities apply the Apple HIG principles to **NayaPay** (a Fintech/Banking application).

Task 1: HIG Analysis

Activity: Open NayaPay and identify 2 strong and 2 weak points based on HIG.

- *Potential Strong Point (Clarity):* The home screen clearly separates "Send Money" and "Add Money" with distinct icons.
- *Potential Weak Point (Deference):* If the promotional banners are too large, they may compete with the core banking content.

Task 2: Gesture Design (Direct Manipulation)

Activity: Design a process to add Rs. 10,000 using gestures only (no text).

- *Concept:* Dragging a coin icon from a "Bank" wallet into the "NayaPay" wallet. The value could be set by a scroll wheel or a vertical slider. This relies on **Direct Manipulation** rather than typing numbers.

Task 3: Security Aesthetics (Aesthetic Integrity)

Activity: Block a misplaced card.

- *Application:* The "Freeze Card" switch should be prominent. The design should look secure—perhaps turning the card visual "grey" or "frozen" immediately upon toggling. This visual change reinforces the **Aesthetic Integrity** of the app's behavior (safety).

Task 4: Transaction Feedback (Feedback & Responsiveness)

Activity: Confirming a sent transaction of Rs. 3,000.

- *Application:* Even if the SMS didn't arrive, the app should show a distinct "Success" checkmark animation and a haptic vibration. The transaction history should instantly update with the new balance. This is critical **Feedback**.

Task 5: Profile Updates (Deference)

Activity: Updating a phone number.

- *Application:* The input field should be clear. The keyboard should automatically switch to the number pad. The UI should defer to the user's goal (typing) by removing unnecessary navigation bars while in "Edit Mode."

Task 6: Bill Payment (Standard Components)

Activity: Paying an electricity bill.

- *Application:* Use the standard iOS **Picker** to select the electricity provider (e.g., K-Electric). Use a standard **Action Sheet** to confirm payment. Using standard components builds trust in financial transactions.

Task 7: Customer Support (Standard Components)

Activity: Contacting support for a failed transaction.

- *Application:* A standard iOS "Compose Message" view or a native "Call" prompt. Do not invent a custom chat interface if a standard one works better.

Task 8: Sending a Gift (Depth & Clarity)

Activity: Sending a wedding gift envelope (Rs. 5000).

- *Application:* Use **Depth**. When selecting "Send Gift," a specialized envelope UI could slide up (modal view) over the dashboard. The animation of the envelope closing and flying off adds delight and confirms the action.

Task 9: Challan Payment (Context)

Activity: Paying a traffic challan.

- *Application:* **Clarity** is vital here. The challan ID input must be distinct. The app could potentially use the camera (Input Interface) to scan the barcode on the physical challan ticket, saving the user from typing errors.